

计算机网络(H) Lab 6 实验报告

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Part 1

任务要求

本实验要求实现分布式异步距离向量路由算法：**初始化 (rtinit)**: 每个节点初始化其距离表，反映与直接邻居的连接成本，并将自己的最小成本向量发送给所有直接邻居。**更新 (rtupdate)**: 当收到邻居的路由更新分组时，利用 Bellman-Ford 方程更新本地距离表。 $D_x(y) = \min_v \{ c(x,v) + D_v(y) \}$ 如果到达某个目的地的最小开销发生变化，则将新的最小开销向量发送给所有直接邻居。**打印输出**: 在初始化和更新时，打印当前时间、发送/接收方信息以及当前的距离表内容。

代码实现

rtinit: 初始化距离表 `dist_table` 为无穷，更新 `min_costs`同时向所有直接邻居发送包含自己最小开销向量 (`min_costs`) 的路由分组

rtupdate: 当收到邻居发送的路由分组时，根据 Bellman-Ford 方程更新距离表

```
dist_table.costs[i][sender_id] =
rcvdpkt->mincost[i] + dist_table.costs[sender_id][NODE_ID]
```

这里`dist_table.costs[sender_id][NODE_ID]`存储的是`c(x, sender_id)`

这之后，遍历所有邻居，找到到达每个目的节点的最小开销，并更新 `min_costs`，如果计算出的最小开销与之前的`min_costs`不同，说明路由发生了变化，需要通告邻居，即构造新的路由分组，包含更新后的`min_costs`

其他三个节点的实现逻辑是一致的，，只需要修改`link_cost`，就不赘述了

结果与分析

运行仿真程序，设置 TRACE=2，观察路由收敛过程。最终各节点的距离表收敛如下（已经在output.pdf里圈出来）：

Node 0 最终距离表 (t=9.320):

		via		
D0		1	2	3
----- -----				
	1	1	4	10
dest	2	2	3	9
	3	4	5	7

Node 1 最终距离表 (t=8.227):

		via	
D1		0	2
----- -----			
	0	1	3
dest	2	3	1
	3	5	3

Node 2 最终距离表 (t=17.443):

		via		
D2		0	1	3
----- -----				
	0	3	2	6
dest	1	4	1	5
	3	7	4	2

Node 3 最终距离表 (t=6.969):

		via	
D3		0	2
----- -----			
	0	7	4
dest	1	8	3
	2	9	2

最终最小开销:

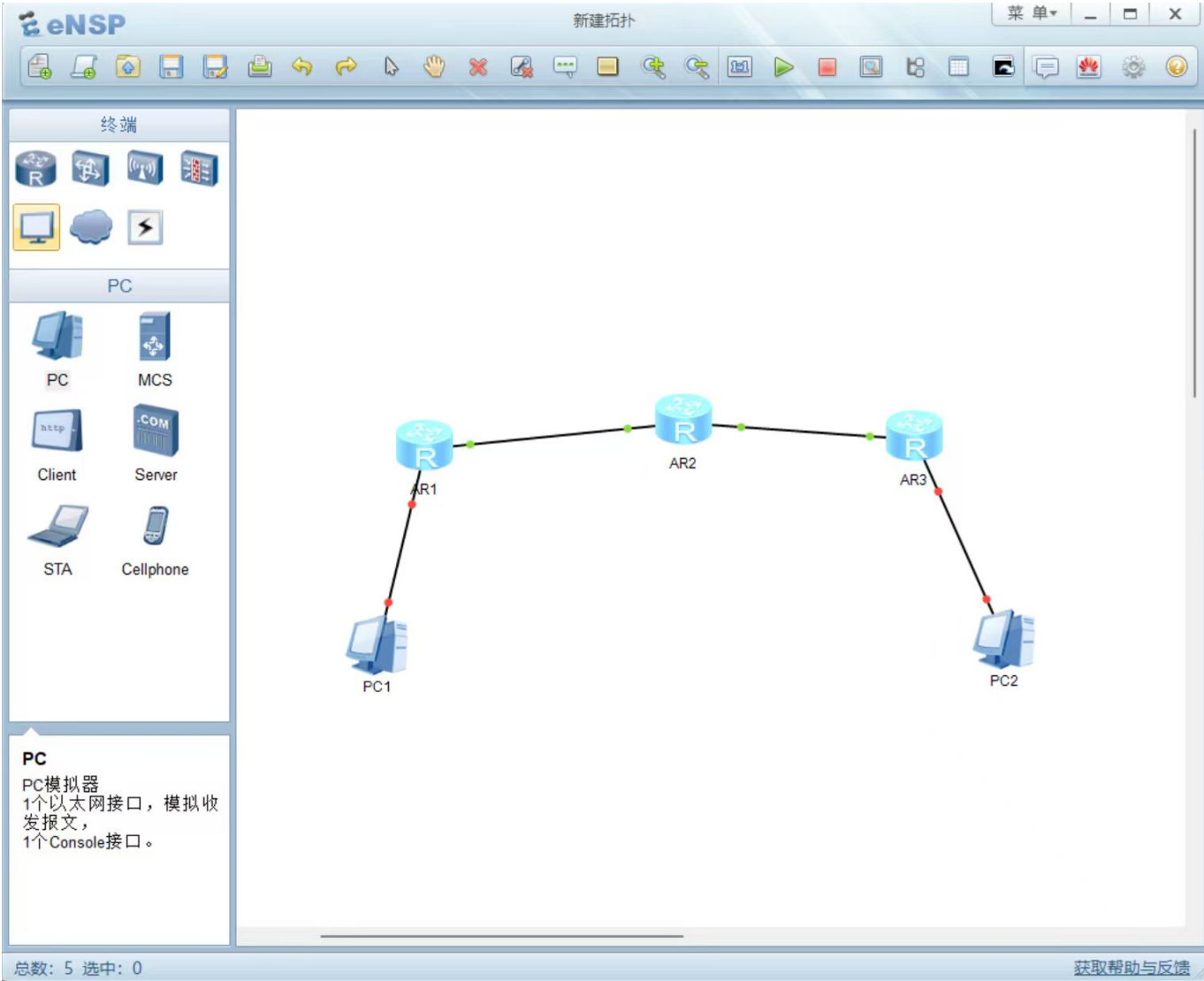
源节点	目的节点	最小开销	Next Hop
Node 0	Node 1	1	Node 1
	Node 2	2	Node 1
	Node 3	4	Node 1
Node 1	Node 0	1	Node 0
	Node 2	1	Node 2
	Node 3	3	Node 2
Node 2	Node 0	2	Node 1
	Node 1	1	Node 1
	Node 3	2	Node 3

源节点	目的节点	最小开销	Next Hop
Node 3	Node 0	4	Node 2
	Node 1	3	Node 2
	Node 2	2	Node 2

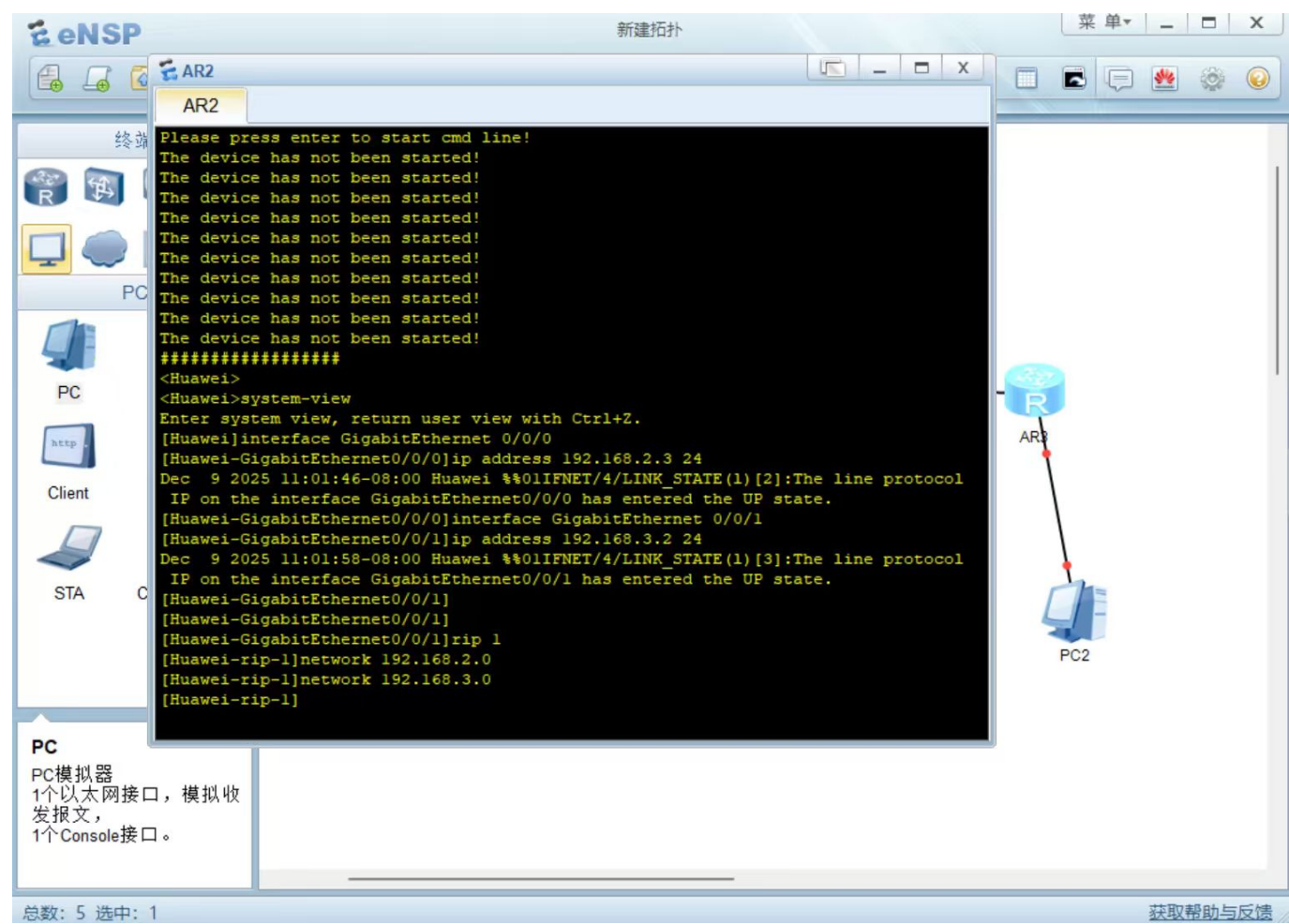
Part 2

任务一：RIP协议报文观察

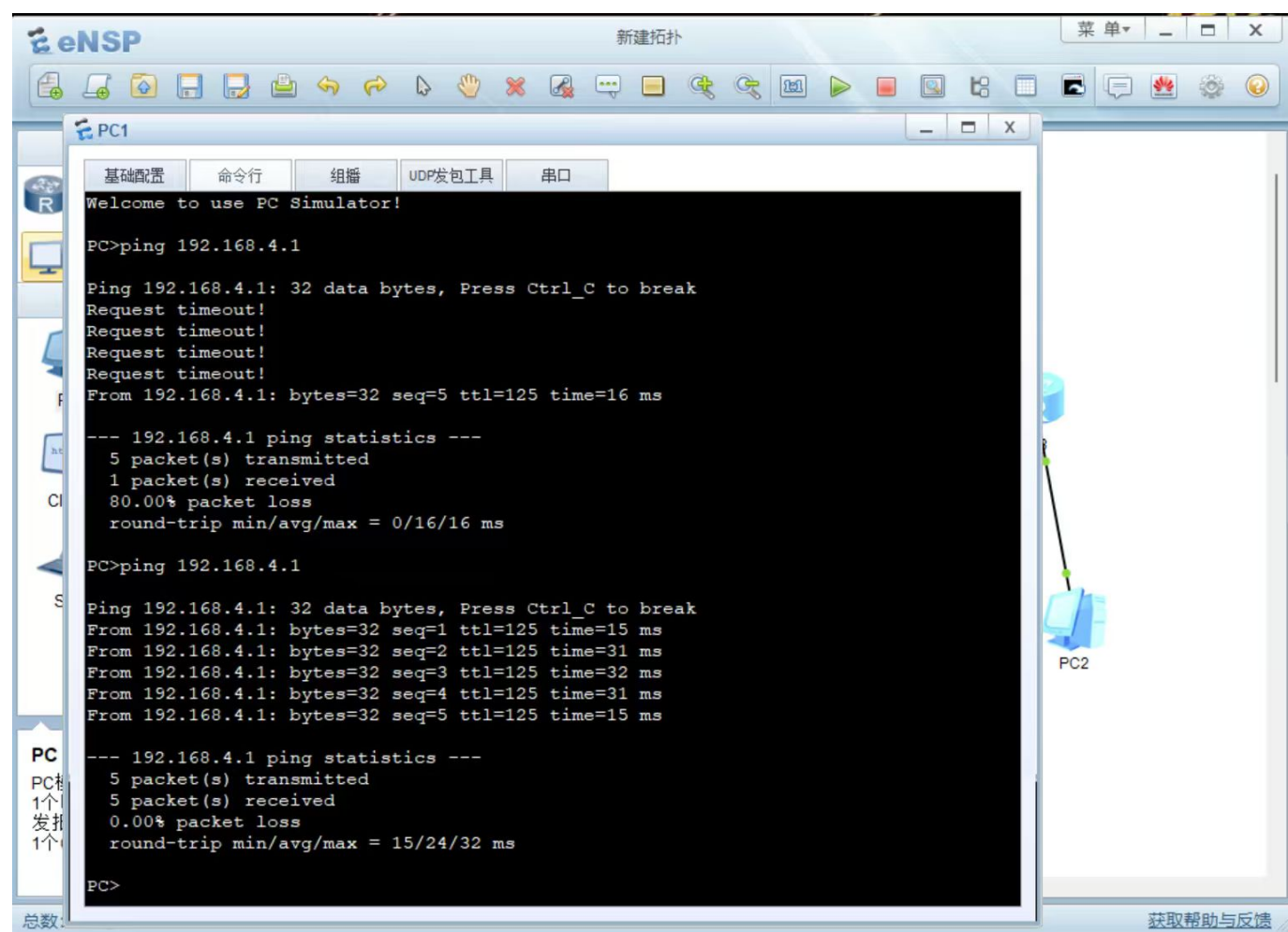
配置拓扑效果：



之后依次完成AR的设置：



按照安装包配置的eNSP，其实和作业网址上提供的有些区别，比如sys命令和int命令就是无效的，但是其他命令直接使用，效果也没有问题，配置好之后可以直接ping通：



得到的三张路由器内IP路由表如下：

eNSP 新建拓扑

终端

PC

PC 模拟器
1个以太网接口，模拟收发报文，
1个Console接口。

AR1

```
<Huawei>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 12      Routes : 12

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
127.0.0.0/8         Direct  0    0       D  127.0.0.1       InLoopBack0
127.0.0.1/32        Direct  0    0       D  127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0       D  127.0.0.1       InLoopBack0
192.168.1.0/24       Direct  0    0       D  192.168.1.254   GigabitEthernet
0/0/0
192.168.1.254/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
192.168.1.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
192.168.2.0/24       Direct  0    0       D  192.168.2.2     GigabitEthernet
0/0/1
192.168.2.2/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
192.168.2.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
192.168.3.0/24       RIP     100  1       D  192.168.2.3     GigabitEthernet
0/0/1
192.168.4.0/24       RIP     100  2       D  192.168.2.3     GigabitEthernet
0/0/1
255.255.255.255/32  Direct  0    0       D  127.0.0.1       InLoopBack0

<Huawei>
```

总数: 5 选中: 1

获取帮助与反馈

eNSP 新建拓扑

终端

PC

PC 模拟器
1个以太网接口，模拟收发报文，
1个Console接口。

AR2

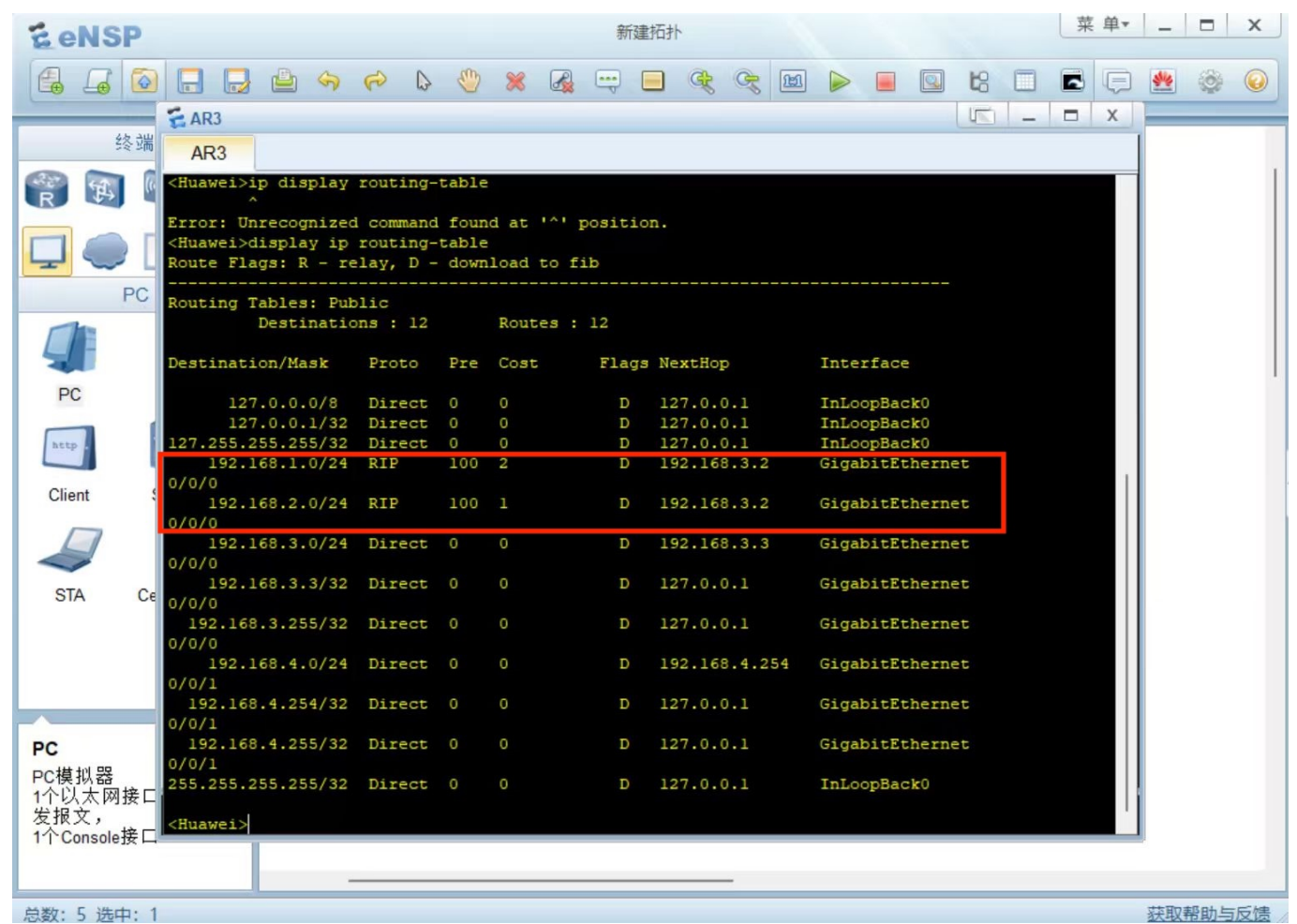
```
<Huawei>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
Destinations : 12      Routes : 12

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
127.0.0.0/8         Direct  0    0       D  127.0.0.1       InLoopBack0
127.0.0.1/32        Direct  0    0       D  127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0       D  127.0.0.1       InLoopBack0
192.168.1.0/24       RIP     100  1       D  192.168.2.2     GigabitEthernet
0/0/0
192.168.2.0/24       Direct  0    0       D  192.168.2.3     GigabitEthernet
0/0/0
192.168.2.3/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
192.168.2.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/0
192.168.3.0/24       Direct  0    0       D  192.168.3.2     GigabitEthernet
0/0/1
192.168.3.2/32      Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
192.168.3.255/32    Direct  0    0       D  127.0.0.1       GigabitEthernet
0/0/1
192.168.4.0/24       RIP     100  1       D  192.168.3.3     GigabitEthernet
0/0/1
255.255.255.255/32  Direct  0    0       D  127.0.0.1       InLoopBack0

<Huawei>
```

总数: 5 选中: 1

获取帮助与反馈



抓包出来的截图如下：

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
2	0.000000	255.255.255.255	192.168.1.254	ICMP	70	Destination unreachable (Port unreachable)
3	29.016000	192.168.1.254	255.255.255.255	RIPv1	146	Response
4	29.016000	255.255.255.255	192.168.1.254	ICMP	70	Destination unreachable (Port unreachable)
5	54.063000	192.168.1.254	255.255.255.255	RIPv1	146	Response
6	54.063000	255.255.255.255	192.168.1.254	ICMP	70	Destination unreachable (Port unreachable)
7	80.078000	192.168.1.254	255.255.255.255	RIPv1	146	Response
8	80.078000	255.255.255.255	192.168.1.254	ICMP	70	Destination unreachable (Port unreachable)
9	105.359000	HuaweiTechno_ea:4d:...	Broadcast	ARP	60	Who has 192.168.1.254? Tell 192.168.1.1
10	105.375000	HuaweiTechno_a7:6a:...	HuaweiTechno_ea:4d:...	ARP	60	192.168.1.254 is at 00:e0:fc:a7:6a:e4
11	105.375000	192.168.1.1	192.168.4.1	ICMP	74	Echo (ping) request id=0x19b6, seq=1/256, ttl=128 (no response fo...
12	107.375000	192.168.1.1	192.168.4.1	ICMP	74	Echo (ping) request id=0x1bb6, seq=2/512, ttl=128 (reply in 13)
13	107.391000	192.168.4.1	192.168.1.1	ICMP	74	Echo (ping) reply id=0x1bb6, seq=2/512, ttl=126 (request in 12)
14	108.406000	192.168.1.1	192.168.4.1	ICMP	74	Echo (ping) request id=0x1cb6, seq=3/768, ttl=128 (reply in 15)
15	108.422000	192.168.4.1	192.168.1.1	ICMP	74	Echo (ping) reply id=0x1cb6, seq=3/768, ttl=126 (request in 14)
16	109.422000	192.168.1.1	192.168.4.1	ICMP	74	Echo (ping) request id=0x1db6, seq=4/1024, ttl=128 (reply in 17)
17	109.438000	192.168.4.1	192.168.1.1	ICMP	74	Echo (ping) reply id=0x1db6, seq=4/1024, ttl=126 (request in 16)
18	110.078000	192.168.1.254	255.255.255.255	RIPv1	146	Response
19	110.078000	255.255.255.255	192.168.1.254	ICMP	70	Destination unreachable (Port unreachable)
20	110.438000	192.168.1.1	192.168.4.1	ICMP	74	Echo (ping) request id=0x1eb6, seq=5/1280, ttl=128 (reply in 21)
21	110.453000	192.168.4.1	192.168.1.1	ICMP	74	Echo (ping) reply id=0x1eb6, seq=5/1280, ttl=126 (request in 20)
22	144.109000	192.168.1.254	255.255.255.255	RIPv1	146	Response
23	144.109000	255.255.255.255	192.168.1.254	ICMP	70	Destination unreachable (Port unreachable)

Time to Live: 128
Protocol: ICMP (1)
Header Checksum: 0xbe54 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.1
Destination Address: 192.168.4.1
[Stream index: 1]

Internet Control Message Protocol
Type: 8 (Echo (ping) request)
Code: 0
Checksum: 0x6cc7 [correct]
[Checksum Status: Good]
Identifier (BE): 6582 (0x19b6)
Identifier (LE): 46617 (0xb619)
Sequence Number (BE): 1 (0x0001)
Sequence Number (LE): 256 (0x0100)
[No response seen]

Data (32 bytes)
Data: 08090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f2021222324252627
[Length: 32]

0000 00 e0 fc a7 6a e4
0010 00 3c b6 19 40 00
0020 04 01 08 00 6c c7
0030 0a 0f 10 11 12 13
0040 1e 1f 20 21 22 23

Standard input: <live capture in progress> 分組: 23 配置: Default

具体报文如下：

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.2.2	255.255.255.255	RIPv1	86	Response
2	32.016000	192.168.2.2	255.255.255.255	RIPv1	86	Response
3	64.047000	192.168.2.2	255.255.255.255	RIPv1	86	Response
4	95.047000	192.168.2.2	255.255.255.255	RIPv1	86	Response
5	123.047000	192.168.2.2	255.255.255.255	RIPv1	86	Response
6	154.063000	192.168.2.2	255.255.255.255	RIPv1	86	Response
7	189.078000	192.168.2.2	255.255.255.255	RIPv1	86	Response
8	218.094000	192.168.2.2	255.255.255.255	RIPv1	86	Response
9	246.125000	192.168.2.2	255.255.255.255	RIPv1	86	Response
10	272.125000	192.168.2.2	255.255.255.255	RIPv1	86	Response
11	302.157000	192.168.2.2	255.255.255.255	RIPv1	86	Response
12	327.157000	192.168.2.2	255.255.255.255	RIPv1	86	Response
13	358.172000	192.168.2.2	255.255.255.255	RIPv1	86	Response
14	390.188000	192.168.2.2	255.255.255.255	RIPv1	86	Response
15	415.203000	192.168.2.2	255.255.255.255	RIPv1	86	Response
16	441.235000	192.168.2.2	255.255.255.255	RIPv1	86	Response
17	471.235000	192.168.2.2	255.255.255.255	RIPv1	86	Response
18	498.250000	192.168.2.2	255.255.255.255	RIPv1	86	Response

Frame 1: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0

Section number: 1

Interface id: 0 (-)

Interface name: -

Interface description: Standard input

Encapsulation type: Ethernet (1)

Arrival Time: Jan 1, 1970 08:42:39.968000000 中国标准时间

Standard input: <live capture in progress> 分组: 18 配置: Default

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.2.2	255.255.255.255	RIPv1	86	Response

Frame 1: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface -, id 0

Section number: 1

Interface id: 0 (-)

Interface name: -

Interface description: Standard input

Encapsulation type: Ethernet (1)

Arrival Time: Jan 1, 1970 08:42:39.968000000 中国标准时间

UTC Arrival Time: Jan 1, 1970 00:42:39.968000000 UTC

Epoch Arrival Time: 2559.968000000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.000000000 seconds]

[Time delta from previous displayed frame: 0.000000000 seconds]

[Time since reference or first frame: 0.000000000 seconds]

Frame Number: 1

Frame Length: 86 bytes (688 bits)

Capture Length: 86 bytes (688 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:udp:rip]

[Coloring Rule Name: IPv4 TTL low or unexpected]

[Coloring Rule String: (ip.dst != 224.0.0.0/4 && ip.ttl < 5 && !(pim || ospf || eigrp || bgp || tcp.port==179)) || (ip.dst == 224.0.0.0/4)]

Ethernet II, Src: HuaweiTechno_54:5d:30 (00:e0:fc:54:5d:30), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

...1... = LG bit: Locally administered address (this is NOT the factory default)

...1... = IG bit: Group address (multicast/broadcast)

Source: HuaweiTechno_54:5d:30 (00:e0:fc:54:5d:30)

...0... = LG bit: Globally unique address (factory default)

...0... = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.2.2, Dst: 255.255.255.255

0100... = Version: 4

...0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS6, ECN: Not-ECT)

Standard input: <live capture in progress> 分组: 19 配置: Default

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
20	561.266000	192.168.2.2	255.255.255.255	RIPv1	86	Response

[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.2.2, Dst: 255.255.255.255

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)

Total Length: 72

Identification: 0x0049 (73)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

> Time to Live: 1

Protocol: UDP (17)

Header Checksum: 0xf5f2 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.2.2

Destination Address: 255.255.255.255

[Stream index: 0]

User Datagram Protocol, Src Port: 520, Dst Port: 520

Source Port: 520

Destination Port: 520

Length: 52

Checksum: 0xb273 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

[Stream Packet Number: 1]

> [Timestamps]

[Time since first frame: 0.000000000 seconds]

[Time since previous frame: 0.000000000 seconds]

UDP payload (44 bytes)

Routing Information Protocol

Command: Response (2)

Version: RIPv1 (1)

> IP Address: 192.168.1.0, Metric: 1

Address Family: IP (2)

IP Address: 192.168.1.0

Time to Live (ip.ttl), 1 byte(s)

|| 分组: 20

|| 配置: Default

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
21	588.297000	192.168.2.2	255.255.255.255	RIPv1	86	Response

Identification: 0x0049 (73)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

> Time to Live: 1

Protocol: UDP (17)

Header Checksum: 0xf5f2 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.2.2

Destination Address: 255.255.255.255

[Stream index: 0]

User Datagram Protocol, Src Port: 520, Dst Port: 520

Source Port: 520

Destination Port: 520

Length: 52

Checksum: 0xb273 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

[Stream Packet Number: 1]

> [Timestamps]

[Time since first frame: 0.000000000 seconds]

[Time since previous frame: 0.000000000 seconds]

UDP payload (44 bytes)

Routing Information Protocol

Command: Response (2)

Version: RIPv1 (1)

> IP Address: 192.168.1.0, Metric: 1

Address Family: IP (2)

IP Address: 192.168.1.0

Metric: 1

> IP Address: 192.168.2.0, Metric: 1

Address Family: IP (2)

IP Address: 192.168.2.0

Metric: 1

Time to Live (ip.ttl), 1 byte(s)

|| 分组: 21

|| 配置: Default

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问题1:

在图中指出网络中共有几个子网，每个子网的地址及掩码是什么？

4 个子网，具体而言：

PC1 – AR1 子网，地址192.168.1.0/24 子网掩码255.255.255.0

AR1 – AR2 子网，地址192.168.2.0/24 子网掩码255.255.255.0

AR2 – AR3 子网，地址192.168.3.0/24 子网掩码255.255.255.0

AR3 – PC2 子网，地址192.168.4.0/24 子网掩码255.255.255.0

问题2见上述截图

问题3:

截图举例说明报文各字段的含义。并说明路由器中的IP路由表中的RIP路由表项是如何由RIP协议报文中的距离向量计算得出的

第一张截图中抓到了一系列RIPv1 Response报文 Source192.168.2.2 发送 RIP 更新的路由器接口地址

Destination255.255.255.255 目的地址是 IPv4 的本地广播地址 ProtocolRIPv1 InfoResponse表明 RIP 携带距离向量，不是路由请求 Time1.000000、32... 之间间隔大约30s，也符合 RIP周期性发送距离向量更新

在第二张、第三张截图中可见报文包括各个字段，Ethernet II 首部Destination: Broadcast (ff:ff:ff:ff:ff:ff) 目的地址是广播地址，说明这个 RIP 更新会在该交换网络上被所有设备接收 Type: IPv4 (0x0800)说明上层协议为 IPv4，Total Length可以看到整个 IP 报文长度72 字节

更底层，可以看到IP 负载是一个 UDP 报文，协议号 17

Time to Live (TTL): 1，表示该报文在 IP 层不会被路由转发，只在本地子网内传播

更底层，UDP 首部Source Port: 520, Destination Port: 520 说明RIP 默认使用 UDP 520 端口进行路由信息交换

在第四张截图中，Routing Information Protocol Command: Response (2)表示响应报文，携带距离向量

多条路由项（RTE，Route Table Entry）IP Address: 192.168.1.0, Metric: 1可见该路由器到达该网络的跳数为 1（即直连） 第二条路由项同理，表示源路由器认为到 192.168.2.0 这个网络的代价为 1 跳即 这个 RIPv1 Response 报文里面携带两条路由信息

路由表如何由距离向量计算得出： AR1原本包括两个邻居， 192.168.1.0/24（PC1） 192.168.2.0/24（AR2） 对于直连网，RIP 视为metric = 1

更新时，AR1 接收到来自 AR2 的距离向量，AR2到 192.168.1.0/24 的代价是 1，到 192.168.2.0/24 的代价是 1 之后，AR1 计算经由 AR2 到该网段的距离， 之后，AR1 比较所有可能的邻居路径，取最小值， 因此，AR1 会保持直连路径为最优，不把经 AR2 的 2 跳路径写进路由表

对于不直连的网络，计算方式一样

最终结果以 **R** 标记的 RIP 路由项的形式写入 AR1 路由表中

问题4:

RIP协议中，距离向量的传递是在邻居之间，还是距离向量会直接传播到全网？传递的距离向量是完整的距离向量吗？

问题4

首先，传递范围是邻居之间逐跳传递 **Time to Live: 1**表示RIP 报文在 IP 层不允许被转发；而且目的地址 **255.255.255.255**为本地广播，可见RIP 报文只被广播一次，只能被直接相邻的路由器收到

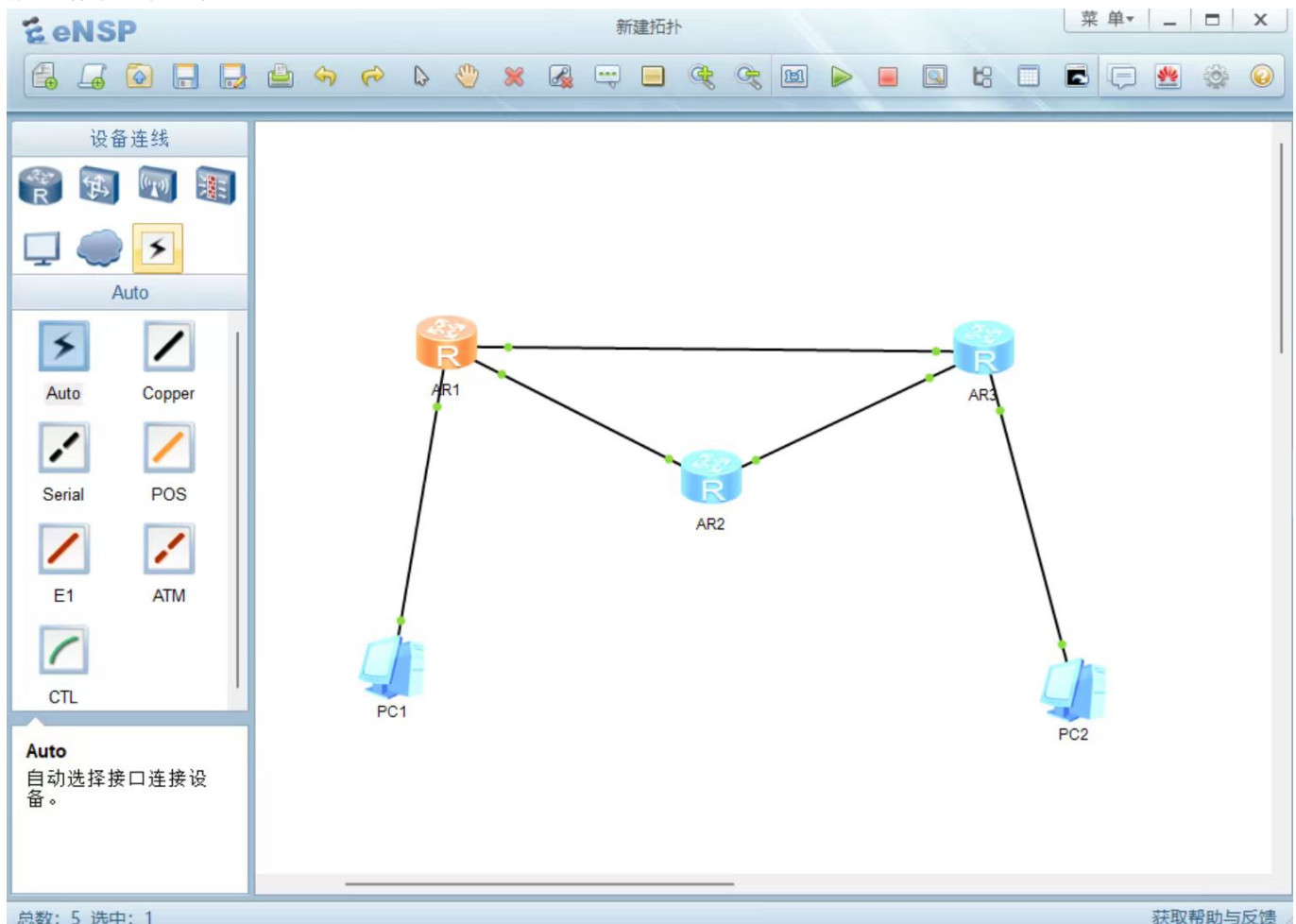
因此，距离向量仅在邻居之间交换

之后，每一台路由器收到邻居的距离向量之后，会更新自己的路由表，再在下一轮 30s 周期性更新中把修正后的距离向量发给自己的邻居去迭代

第四张截图可见单个 RIPv1 Response 报文就包含多条路由项，可见路由器在发送 RIP 更新时，会把自己当前已知的目的网络及其度量值一起全部发送给邻居

第二部分:

配置拓扑效果如下：



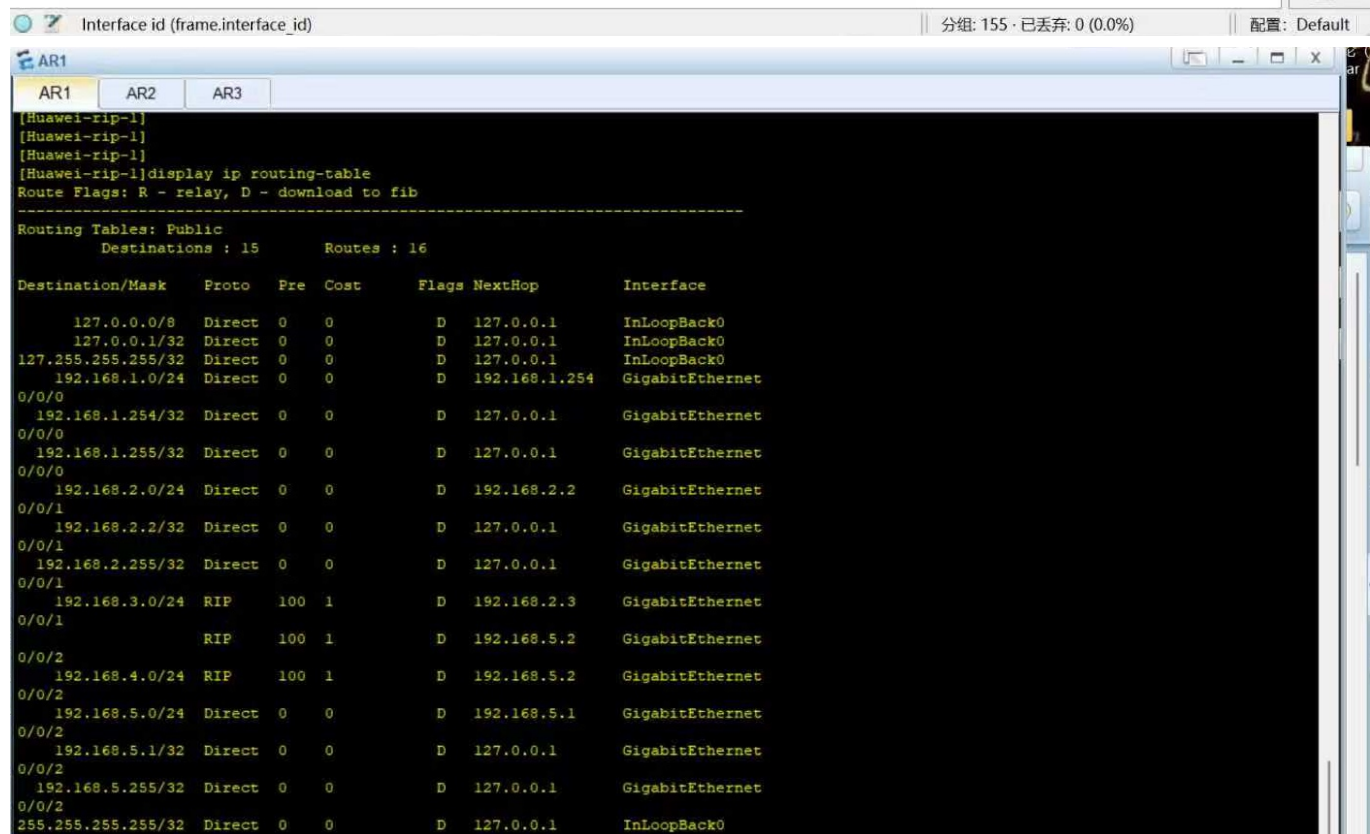
问题1:

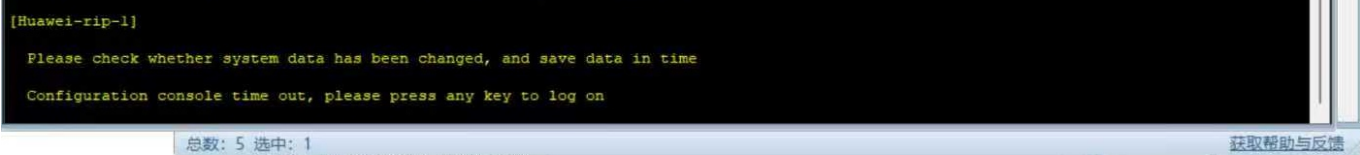
按照要求搭建新的拓扑，完成相关路由器的配置（AR1和AR3的新端口的配置、RIP路由协议的配置）。并在图中指出网络中共有几个子网，每个子网的IP及掩码是什么？

现在是5个子网，具体而言：PC1 - AR1 子网，地址192.168.1.0/24，掩码255.255.255.0 AR1 - AR3 子网，地址192.168.2.0/24，掩码255.255.255.0 AR1 - AR2 子网，地址192.168.3.0/24，掩码255.255.255.0 AR3 - PC2 子网，地址192.168.4.0/24，掩码255.255.255.0 AR2 - AR3 子网，地址192.168.5.0/24，掩码255.255.255.0

问题2:

在各个路由器端口进行抓包，说明在改变网络拓扑后RIP报文中距离向量的变化，以及该变化如何影响IP路由表





AR2

AR1AR2AR3

0.00% packet loss
round-trip min/avg/max = 20/22/30 ms

[Huawei-GigabitEthernet0/0/1]rip 1
[Huawei-rip-1]network 192.168.2.0
[Huawei-rip-1]network 192.168.3.0
[Huawei-rip-1]
[Huawei-rip-1]display ip routing-table
Route Flags: R - relay, D - download to fib

Routing Tables: Public
Destinations : 13Routes : 14

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	RIP	100	1	D	192.168.2.2	GigabitEthernet
0/0/0						
192.168.2.0/24	Direct	0	0	D	192.168.2.3	GigabitEthernet
0/0/0						
192.168.2.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
192.168.2.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
192.168.3.0/24	Direct	0	0	D	192.168.3.2	GigabitEthernet
0/0/1						
192.168.3.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
192.168.3.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
192.168.4.0/24	RIP	100	1	D	192.168.3.3	GigabitEthernet
0/0/1						
192.168.5.0/24	RIP	100	1	D	192.168.2.2	GigabitEthernet
0/0/0						
	RIP	100	1	D	192.168.3.3	GigabitEthernet
0/0/1						
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[Huawei-rip-1]

Please check whether system data has been changed, and save data in time
Configuration console time out, please press any key to log on

总数: 5 选中: 1

获取帮助与反馈

eNSP

新建拓扑

菜单

AR3

AR1AR2AR3

Route Flags: R - relay, D - download to fib

Routing Tables: Public
Destinations : 15Routes : 16

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	RIP	100	1	D	192.168.5.1	GigabitEthernet
0/0/0						
192.168.2.0/24	RIP	100	1	D	192.168.3.2	GigabitEthernet
0/0/1						
	RIP	100	1	D	192.168.5.1	GigabitEthernet
0/0/0						
192.168.3.0/24	Direct	0	0	D	192.168.3.3	GigabitEthernet
0/0/1						
192.168.3.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
192.168.3.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/1						
192.168.4.0/24	Direct	0	0	D	192.168.4.254	GigabitEthernet
0/0/2						
192.168.4.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
192.168.4.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/2						
192.168.5.0/24	Direct	0	0	D	192.168.5.2	GigabitEthernet
0/0/0						
192.168.5.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
192.168.5.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet
0/0/0						
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[Huawei-rip-1]

总数: 5 选中: 1

获取帮助与反馈

修改拓扑，增加 AR1-AR3 直连链路后，原来的路径PC1 - AR1 - AR2 - AR3 - PC2共经过 3 个跳，而新的路径PC1 - AR1 - AR3 - PC2只经过 2 跳

因此，链路建立后，AR3 会在发送给 AR1 的 RIP Response 中更新cost并不断传播下去 最后一张截图可见 Source: 192.168.2.3, Destination: 255.255.255.255, Protocol: RIPv1, Info: 86 Response 说明现在从192.168.2.3 AR3在向本子网广播 RIP Response，即AR1 的距离向量更新时会同时考虑来自 AR2 和 AR3

在 AR3 发出的 RIP Response 中，对网络 192.168.1.0/24 的 metric 更小，因此AR3 的 display ip routing-table可见： 192.168.1.0/24 RIP Cost 1 NextHop 192.168.5.1 Interface GigabitEthernet ... 192.168.2.0/24 RIP Cost 1 NextHop 192.168.3.2 Interface GigabitEthernet ...

即从 AR3 到 AR1/PC1 的最短路径变化了

对于AR1 的 display ip routing-table，直连网络也新增了192.168.2.0/24 Direct ...

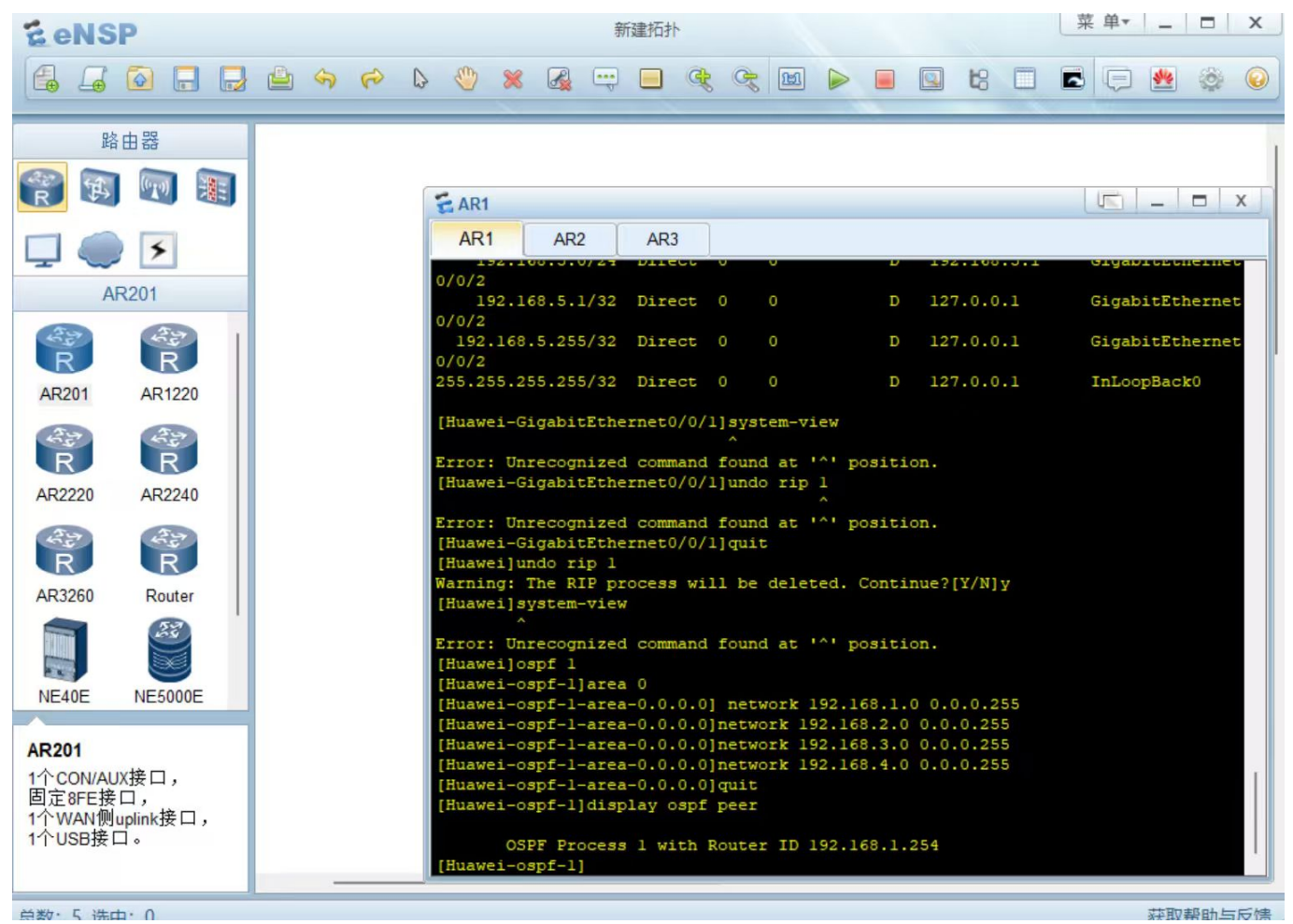
对于PC2 所在的 192.168.4.0/24，AR1 的 RIP 路由表项原本是192.168.4.0/24 的 metric 为3，现在减到2

AR2 的 display ip routing-table可见： 192.168.1.0/24 RIP Cost 1 NextHop 192.168.2.2 192.168.4.0/24 RIP Cost 1 NextHop 192.168.3.3 192.168.2.0/24、192.168.3.0/24 等是 Direct的

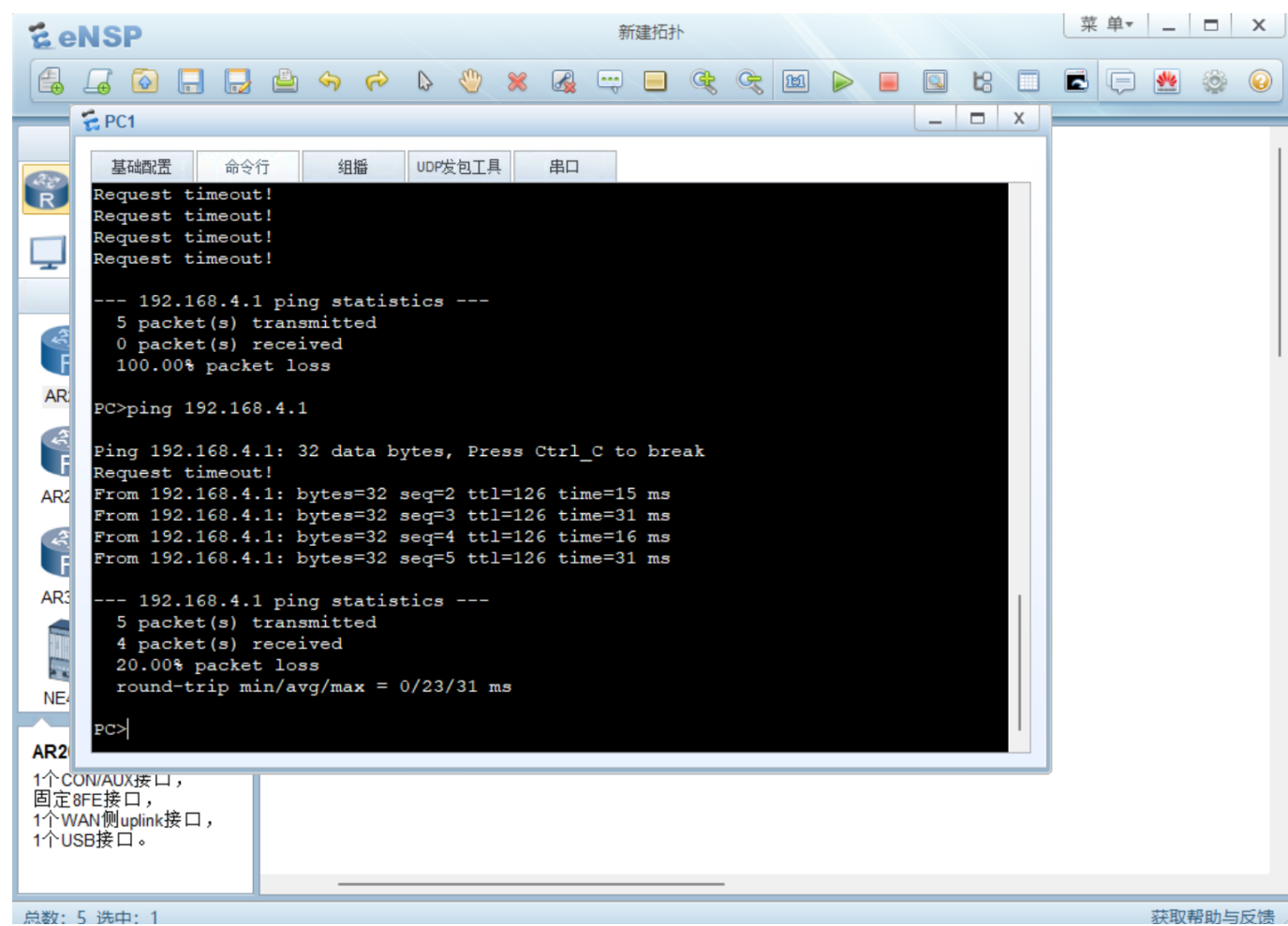
总而言之，最后AR1 到 PC2 网 192.168.4.0/24 的 metric 由 3 跳变为 2 跳，下一跳从 AR2 换成了 AR3；AR3 到 PC1 网 192.168.1.0/24 的 metric 同样减小； 因此在 display ip routing-table 中可见关目的网的表项协议为 RIP的部分，下一跳地址变更了

任务二：OSPF协议报文观察

OSPF的配置命令比较难调，调了半天，配置效果如下：



也能ping通了:



抓包出来的路由表如下：

alt text

路由器

AR201

AR1220

AR2220

AR2240

AR3260

Router

NE40E

NE5000E

AR201
1个CON/AUX接口，
固定8FE接口，
1个WAN侧uplink接口，
1个USB接口。

总数：5 选中：1

新建拓扑

菜单

AR3

AR1AR2AR3

Authentication Sequence: [0]

[Huawei-ospf-1-area-0.0.0.0]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 15Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	OSPF	10	3	D	192.168.3.2	GigabitEthernet0/0/1
192.168.2.0/24	OSPF	10	2	D	192.168.3.2	GigabitEthernet0/0/1
192.168.3.0/24	Direct	0	0	D	192.168.3.3	GigabitEthernet0/0/1
192.168.3.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
192.168.3.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
192.168.4.0/24	Direct	0	0	D	192.168.4.254	GigabitEthernet0/0/2
192.168.4.254/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
192.168.4.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/2
192.168.5.0/24	Direct	0	0	D	192.168.5.2	GigabitEthernet0/0/0
192.168.5.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
192.168.5.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

新建拓扑

菜单

AR2

AR1AR2AR3

Neighbor is up for 00:00:13

Authentication Sequence: [0]

[Huawei-ospf-1-area-0.0.0.0]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 11Routes : 11

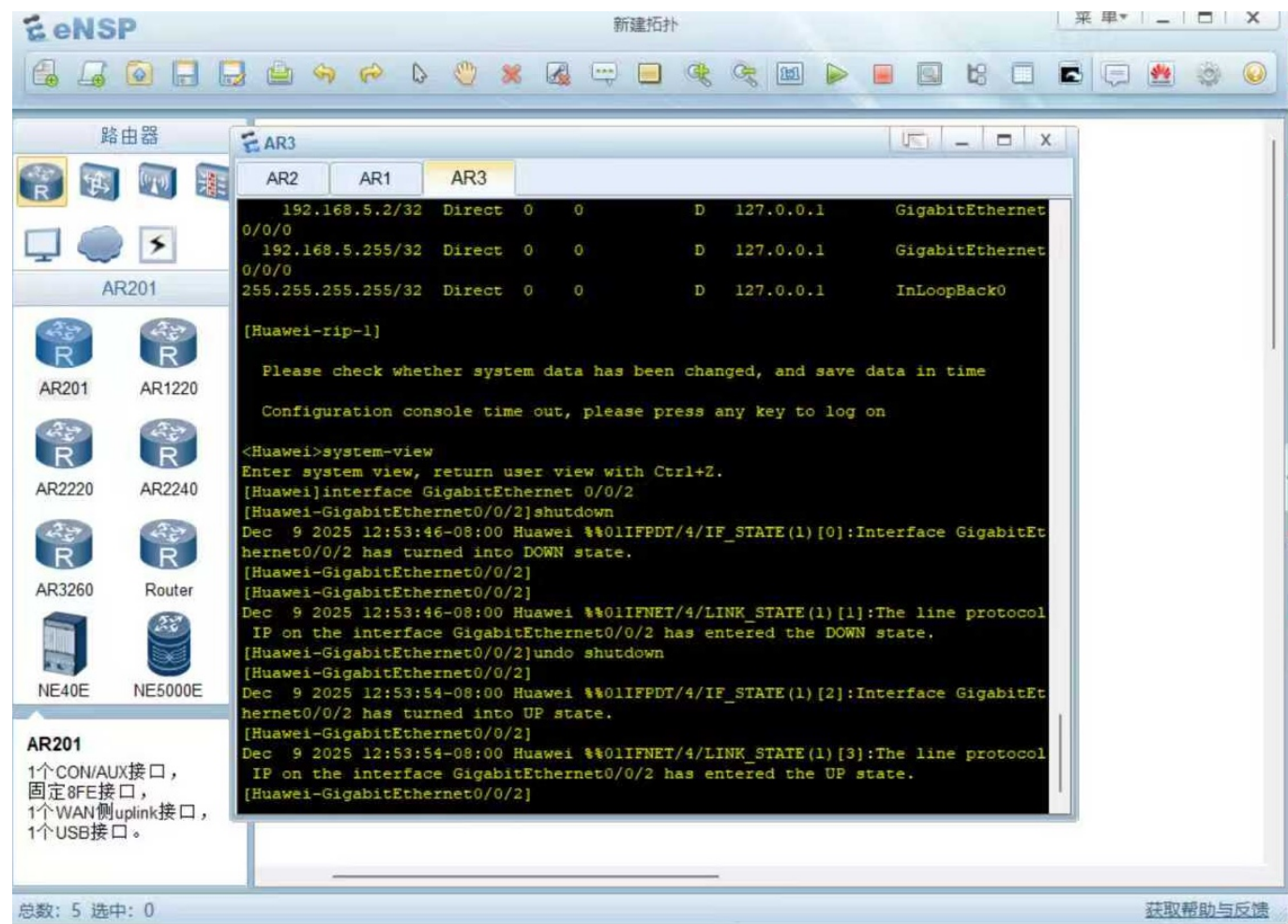
Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
192.168.1.0/24	OSPF	10	2	D	192.168.2.2	GigabitEthernet0/0/0
192.168.2.0/24	Direct	0	0	D	192.168.2.3	GigabitEthernet0/0/0
192.168.2.3/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
192.168.2.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/0
192.168.3.0/24	Direct	0	0	D	192.168.3.2	GigabitEthernet0/0/1
192.168.3.2/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
192.168.3.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet0/0/1
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[Huawei-ospf-1-area-0.0.0.0]

总数：5 选中：1

获取帮助与反馈

之后在抓包过程中也是一直抓不到update，查阅资料了解到每次shutdown一个结点，才能抓到update



抓到的update包如下：

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
33	86.422000	192.168.2.2	192.168.2.3	OSPF	110	LS Update

Frame 33: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface -, id 0

Section number: 1

Interface id: 0 (-)

Interface name: -

Interface description: Standard input

Encapsulation type: Ethernet (1)

Arrival Time: Jan 1, 1970 11:36:16.781000000 中国标准时间

UTC Arrival Time: Jan 1, 1970 03:36:16.781000000 UTC

Epoch Arrival Time: 12976.781000000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.016000000 seconds]

[Time delta from previous displayed frame: 0.016000000 seconds]

[Time since reference or first frame: 86.422000000 seconds]

Frame Number: 33

Frame Length: 110 bytes (880 bits)

Capture Length: 110 bytes (880 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:ospf]

[Coloring Rule Name: OSPF State Change]

[Coloring Rule String: ospf.msg != 1]

Ethernet II, Src: HuaweiTechno_a7:6a:e5 (00:e0:fc:a7:6a:e5), Dst: HuaweiTechno_be:59:31 (00:e0:fc:be:59:31)

Destination: HuaweiTechno_be:59:31 (00:e0:fc:be:59:31)

... .. = LG bit: Globally unique address (factory default)

... .. = IG bit: Individual address (unicast)

Source: HuaweiTechno_a7:6a:e5 (00:e0:fc:a7:6a:e5)

... .. = LG bit: Globally unique address (factory default)

... .. = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 2]

Internet Protocol Version 4, Src: 192.168.2.2, Dst: 192.168.2.3

0100 = Version: 4

... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)

Total Length: 96

Identification: 0x0236 (566)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 1

Protocol: OSPF IGP (89)

Header Checksum: 0x30fa [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.2.2

Destination Address: 192.168.2.3

[Stream index: 2]

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
33	86.422000	192.168.2.2	192.168.2.3	OSPF	110	LS Update

Frame Number: 33

Frame Length: 110 bytes (880 bits)

Capture Length: 110 bytes (880 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:ospf]

[Coloring Rule Name: OSPF State Change]

[Coloring Rule String: ospf.msg != 1]

Ethernet II, Src: HuaweiTechno_a7:6a:e5 (00:e0:fc:a7:6a:e5), Dst: HuaweiTechno_be:59:31 (00:e0:fc:be:59:31)

Destination: HuaweiTechno_be:59:31 (00:e0:fc:be:59:31)

... .. = LG bit: Globally unique address (factory default)

... .. = IG bit: Individual address (unicast)

Source: HuaweiTechno_a7:6a:e5 (00:e0:fc:a7:6a:e5)

... .. = LG bit: Globally unique address (factory default)

... .. = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 2]

Internet Protocol Version 4, Src: 192.168.2.2, Dst: 192.168.2.3

0100 = Version: 4

... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)

Total Length: 96

Identification: 0x0236 (566)

> 000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 1

Protocol: OSPF IGP (89)

Header Checksum: 0x30fa [validation disabled]

[Header checksum status: Unverified]
Source Address: 192.168.2.2
Destination Address: 192.168.2.3
[Stream index: 2]

Open Shortest Path First

OSPF Header

Version: 2
Message Type: LS Update (4)
Packet Length: 76
Source OSPF Router: 192.168.1.254
Area ID: 0.0.0.0 (Backbone)
Checksum: 0x152a [correct]
Auth Type: Null (0)
Auth Data (none): 0000000000000000

LS Update Packet

Number of LSAs: 1

LSA-type 1 (Router-LSA), len 48

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(V) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
32	86.406000	192.168.2.3	192.168.2.2	OSPF	166	DB Description
33	86.422000	192.168.2.2	192.168.2.3	OSPF	110	LS Update
34	86.422000	192.168.2.2	192.168.2.3	OSPF	82	LS Request

... 0 ... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.2.2, Dst: 224.0.0.5

Open Shortest Path First

OSPF Header

Version: 2
Message Type: LS Update (4)
Packet Length: 108
Source OSPF Router: 192.168.1.254
Area ID: 0.0.0.0 (Backbone)
Checksum: 0x778b [correct]
Auth Type: Null (0)
Auth Data (none): 0000000000000000

LS Update Packet

Number of LSAs: 2

LSA-type 2 (Network-LSA), len 32
.000 1110 0001 0000 = LS Age (seconds): 3600
0... .. = Do Not Age Flag: 0
> Options: 0x02, (E) External Routing
LS Type: Network-LSA (2)
Link State ID: 192.168.2.2
Advertising Router: 192.168.1.254
Sequence Number: 0x80000002
Checksum: 0x9416
Length: 32
Netmask: 255.255.255.0
Attached Router: 192.168.1.254
Attached Router: 192.168.2.3

LSA-type 1 (Router-LSA), len 48
.000 0000 0000 0001 = LS Age (seconds): 1
0... .. = Do Not Age Flag: 0
> Options: 0x02, (E) External Routing
LS Type: Router-LSA (1)
Link State ID: 192.168.1.254
Advertising Router: 192.168.1.254
Sequence Number: 0x8000000b
Checksum: 0x42ab
Length: 48
> Flags: 0x00
Number of Links: 2
> Type: Stub ID: 192.168.1.0 Data: 255.255.255.0 Metric: 1
> Type: Transit ID: 192.168.2.3 Data: 192.168.2.2 Metric: 50

Standard input: <live capture in progress>

分组: 65

配置: Default

正在捕获 Standard input

文件(F) 编辑(E) 视图(V) 跳转(G) 捕获(C) 分析(A) 统计(S) 电话(Y) 无线(W) 工具(T) 帮助(H)

应用显示过滤器 ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
32	86.406000	192.168.2.3	192.168.2.2	OSPF	166	DB Description
33	86.422000	192.168.2.2	192.168.2.3	OSPF	110	LS Update
34	86.422000	192.168.2.2	192.168.2.3	OSPF	82	LS Request

Frame 38: 142 bytes on wire (1136 bits), 142 bytes captured (1136 bits) on interface -, id 0

Section number: 1

Interface id: 0 (-)

Interface name: -

Interface description: Standard input

Encapsulation type: Ethernet (1)

Arrival Time: Jan 1, 1970 11:36:16.796000000 中国标准时间

UTC Arrival Time: Jan 1, 1970 03:36:16.796000000 UTC

Epoch Arrival Time: 12976.796000000

[Time shift for this packet: 0.000000000 seconds]

[Time delta from previous captured frame: 0.000000000 seconds]

[Time delta from previous displayed frame: 0.000000000 seconds]

[Time since reference or first frame: 86.437000000 seconds]

Frame Number: 38

Frame Length: 142 bytes (1136 bits)

Capture Length: 142 bytes (1136 bits)

[Frame is marked: False]

[Frame is ignored: False]

[Protocols in frame: eth:ethertype:ip:ospf]

[Coloring Rule Name: OSPF State Change]

[Coloring Rule String: ospf.msg != 1]

Ethernet II, Src: HuaweiTechno_a7:6a:e5 (00:e0:fc:a7:6a:e5), Dst: IPv4mcast_05 (01:00:5e:00:00:05)

Destination: IPv4mcast_05 (01:00:5e:00:00:05)

.....0. = LG bit: Globally unique address (factory default)

.....1. = IG bit: Group address (multicast/broadcast)

Source: HuaweiTechno_a7:6a:e5 (00:e0:fc:a7:6a:e5)

.....0. = LG bit: Globally unique address (factory default)

.....0. = IG bit: Individual address (unicast)

Type: IPv4 (0x0800)

[Stream index: 0]

Internet Protocol Version 4, Src: 192.168.2.2, Dst: 224.0.0.5

Open Shortest Path First

OSPF Header

Version: 2

Message Type: LS Update (4)

Packet Length: 108

Source OSPF Router: 192.168.1.254

Area ID: 0.0.0.0 (Backbone)

Checksum: 0x778b [correct]

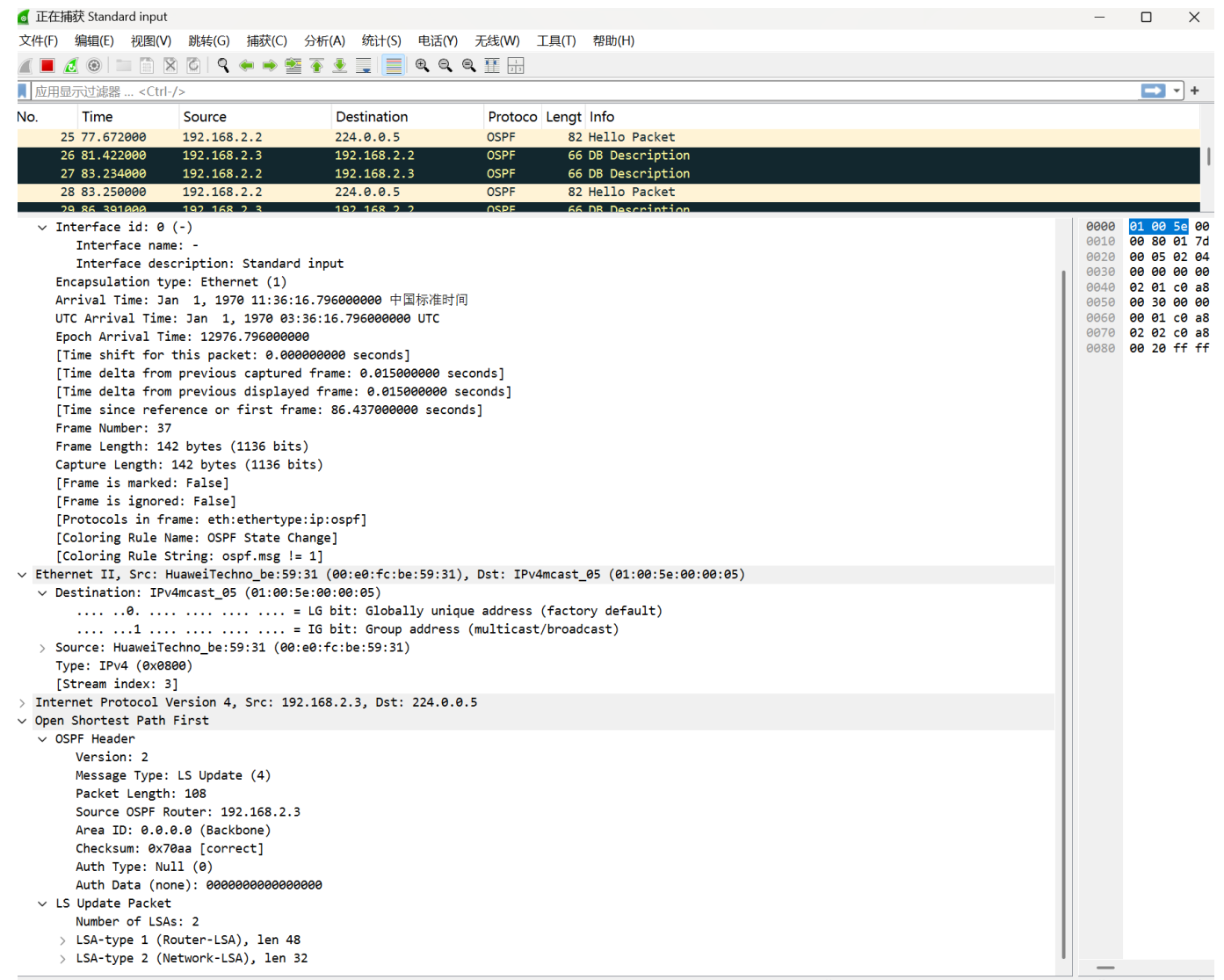
Auth Type: Null (0)

Auth Data (none): 0000000000000000

LS Update Packet

Number of LSAs: 2

Standard input: <live capture in progress> 分组: 64 配置: Default



问题1、2图见上

问题3:

观察OSPF协议中LS Update Packet，截图举例说明Router-LSA和Network-LSA各自的作用与区别。并说明路由器中的IP路由表中的OSPF路由表项是如何由OSPF协议报文中的LSA计算得出的，截图并说明 LS Update 报文（Frame 38）可见Source 192.168.2.2 Destination 224.0.0.5 Protocol: OSPF Info: LS Update

展开 IP 和 OSPF 头部可见：IP目的地址224.0.0.5 发送给本网段所有 OSPF 路由器，TTL: 1即只在本链路上转发

OSPF Header中Version: 2 Message Type: 4 (LS Update)，即是链路状态更新报文 Source OSPF Router: 192.168.1.254，即这条 LSA 是由 Router ID 为 192.168.1.254 的那台路由器产生 Area ID: 0.0.0.0即属于骨干区域

LS Update Packet部分可见Number of LSAs: 2，即这个 LS Update 包里同时携带了两条 LSA，LSA-type: 2 (Network-LSA)，和LSA-type: 1 (Router-LSA)

对于Network-LSA，可见：LSA-type: 2 (Network-LSA) Advertising Router: 192.168.1.254

说明这条 Network-LSA 是由该网段上的DR（指定路由器）产生的，从网络的角度，标记这广播网，之后广播 update信息

对于Router-LSA，可见： LSA-type: 1 (Router-LSA) Number of Links: 2

对于两个link，可见： Link 1为Stub 链路，ID: 192.168.1.0 指向一个非 OSPF 邻居 Data: 255.255.255.0 Metric: 1，即这台路由器直连到网络 192.168.1.0/24，代价为 1

Link 2为Transit 链路，ID: 192.168.2.3 指向广播网络 Data: 192.168.2.2 Metric: 5（也截图到了一个50，应该还是没有完全收敛导致的），即路由器通过192.168.2.2连接到广播网代价为5

总而言之，二者区别在于： Router-LSA（Type 1）是每一台OSPF路由器为自己生成，用来描述自身的link state的， Network-LSA（Type 2）仅由多路访问网络中的DR生成

此外， Router-LSAID等于Router ID，而Network-LSA ID等于该DR所在网络的IP地址，描述网络整体信息

由此，就可以避免冗余信息，

算出路由表中的 OSPF表项的方法：

首先是构建链路状态数据库（LSDB），每台路由器都会不停从邻居收到各种 LSA（包括 Router-LSA 和 Network-LSA，查阅资料可见其实一共有7种LSA），把它们存入本地的 LSDB 中，通过泛洪是的区域内每台路由器的 LSDB 一致

之后，各个路由器从 LSDB 计算拓扑图，再通过SPF 算法（Dijkstra）求最短路径树

例如上述截图可见AR1 的路由表中有 192.168.1.0/24 Direct 192.168.3.0/24 OSPF ... NextHop 192.168.2.3 可见 AR1 运行 SPF 得出 AR1 - AR3 - 192.168.3.0/24 为代价最小路径进而产生对应的 OSPF 路由

*AR2 的路由表可见192.168.1.0/24 OSPF Cost 2 NextHop 192.168.2.2，也是计算得到，到达AR1 网络再到192.168.1.0/24 的Cost 为2

问题4:

OSPF协议中的LSA信息传递是在邻居间交换还是在全网传递，截图并说明

LS Update 报文里有两种情况，首先是单播到邻居 的 LS Update，图2可见： Source: 192.168.2.2 Destination: 192.168.2.3 OSPF, LS Update IP TTL=1

另一种为组播到 AllSPFRouters(224.0.0.5) 的 LS Update，图5可见： 目的 MAC: 01:00:5e:00:00:05 (IPv4mcast_05) 目的 IP: 224.0.0.5 TTL也是 1

可见，OSPF 的控制报文（包括 LSA）TTL=1，并且目的地址为图上邻居IP*或者链路组播地址 224.0.0.5/6

即，单个 LS Update 包只在本链接的邻居之间传递，之后每个邻居收到后会继续泛洪，最后实现是整个区域内传播

截图可见AR1 的 LAN 192.168.1.0/24 的直连信息，是 AR1 在自己的 Router-LSA 给出的，但在 AR2、AR3 的路由表（虽然他们不直连）也有，可见经过邻居转发最后link state使得整个网络的拓扑是每个路由器都维护了的，即通过中转的泛洪转发，一跳一跳实现的

附录

一、参考

距离向量 <https://www.geeksforgeeks.org/computer-networks/distance-vector-routing-dvr-protocol/>
https://eng.libretexts.org/Bookshelves/Computer_Science/Networks/An_Introduction_to_Computer_Networks_%28Dordal%29/09%3A_Routing-Update_Algorithms/9.02%3A_Distance-Vector_Routing-Update_Algorithm https://gaia.cs.umass.edu/kurose_ross/interactive/disVector.php

代码实现 <https://github.com/jdaniels539/Distance-Vector-Routing-Algorithm/blob/master/proj3.c>
<https://ankurm.com/implementation-of-distance-vector-routing-dvr-algorithm-in-c-2/>

OSPF-LSA <https://zhuanlan.zhihu.com/p/463267606> <https://www.cnblogs.com/lwmyyyy/articles/18804335>

二、代码编写、配环境经验

prog3.c疑似是上世纪的产物，还在使用K&R语法和gnu89；首先，新开的变量放最前；使用**-std=gnu89**编译选项；添加**-Wno-return-mismatch**编译选项，避免关于**tolayer2函数**的报错（该函数在代码中定义为void 类型，但遗留调用方隐式要求 int 类型）

具体的编译指令，在Makefile里

Part 2配置eNSP非常复杂，重点在于不要改动wireshark，用ver 4的wireshark，不能用网盘里的ver 3；另外，老的dll务必要备份保留，我的电脑配着配着就黑屏了，幸好留了一手，dll文件备份后可以复原

不得不说，eNSP非常离谱，不仅配置非常繁琐，而且开始抓包之后电脑截屏功能有时候无法使用（原因未知，但是确定和eNSP相关），不得不使用其他软件自带工具来截屏，可能有些模糊，望理解